

Homework 1

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1. Compute $G(k)$ for $k = 110001$

$$G(110001) = 11110000110$$

$$\begin{array}{lcl} 2. & m & = 001110011000 \\ & c & = 11001001110 \end{array}$$

x	y	$x \oplus y$
0	0	0
0	1	1
1	0	1
1	1	0

3. We need to calculate the probability that a random bit $G_i(k) = 0$ for $i \leq 2n$, where $G(k)$ is of length $2n$.

From the structure of $G(k)$, each bit in the original key k appears twice, so the probability of a 0-bit in $G(k)$ depends on the proportion of 0s in k .

In $k = 110001$, there are 3 zero bits and 3 one bits. Since each bit appears twice in $G(k)$, the probability of any given bit being 0 is:

$$P(G_i(k) = 0) = \frac{\text{number of 0's in } k}{n} = \frac{3}{6} = \frac{1}{2} = 0.5$$

This proves that PRG is not secure because there is a 50% chance of generating a 0 or 1 and that does not imply security.

$$4. P(A(r_1, \dots, r_{2n}) = 0) = \frac{1}{2}$$

$$P(A(G(k_1, \dots, k_n)) = 0) = \frac{1}{2}$$

$$\left| \frac{1}{2} - \frac{1}{2} \right| = 0$$

The advantage of $A = 0$.

5. $P(A(r_1, \dots, r_{2n}) = 0) = \frac{1}{2}$

$$p(\vee A(G(k_1, \dots, k_n) = 0) = 1$$

$$\frac{1-1}{2} = \frac{1}{2} = 0.5$$

the advantage of $A = 0.5$

6. a. $G(k_1, \dots, k_m) = k_1 k_2 k_2 k_1 k_3 k_4 k_4 k_3 k_5 k_6 k_6 k_5 \dots$
 $\quad\quad\quad m_0 \quad\quad\quad = 0 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1$
 $\quad\quad\quad C \quad\quad\quad = k_1 k_2 \overline{k_2} \overline{k_1} k_3 k_4 \overline{k_4} \overline{k_3} k_5 k_6 \overline{k_6} \overline{k_5}$

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graph LR
    c1((c1)) --- AND[AND]
    c4((c4)) --- AND
    AND --- NOT1[NOT]
    NOT1 --- NOT2[NOT]
  
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$$C_1 \neq C_4, \therefore$$

$P(W_0) = 1$, the C_1 and C_u will always be different

b. $G(k_1, \dots, k_n) = k_1 k_2 k_2 k_3 k_4 k_4 k_3 k_5 k_6 k_6 k_5$
 $m_1 = 0000000000$
 $C = k_1 k_2 k_2 k_3 k_4 k_4 k_3 k_5 k_6 k_6 k_5$
 $G_1 \nearrow \quad \nwarrow C_4$

$$C_1 = C_4 \therefore$$

$P(w_i) = 0, c,$ and c_y will always be same.

c. Advantage = $|P(w_0) - P(w_1)|$

$$= |1 - 0| = 1 \leftarrow \text{biggest value.}$$